

```

pvecback[pba->index_bg_H] = sqrt(rho_tot-pba->K/a/a);

/** - compute derivative of H with respect to conformal time */
pvecback[pba->index_bg_H_prime] = - (3./2.) * (rho_tot + p_tot) * a + pba->K/a;

```

Compute the density
and pressure for
each component and
add it to the total
density and pressure.

```

/* photons */
pvecback[pba->index_bg_rho_g] = pba->Omega0_g * pow(pba->H0,2) / pow(a,4);
rho_tot += pvecback[pba->index_bg_rho_g];
p_tot += (1./3.) * pvecback[pba->index_bg_rho_g];
dp_dloga += -(4./3.) * pvecback[pba->index_bg_rho_g];
rho_r += pvecback[pba->index_bg_rho_g];

/* baryons */
pvecback[pba->index_bg_rho_b] = pba->Omega0_b * pow(pba->H0,2) / pow(a,3);
rho_tot += pvecback[pba->index_bg_rho_b];
p_tot += 0;
rho_m += pvecback[pba->index_bg_rho_b];

/* cdm */
if (pba->has_cdm == _TRUE_) {
    pvecback[pba->index_bg_rho_cdm] = pba->Omega0_cdm * pow(pba->H0,2) / pow(a,3);
    rho_tot += pvecback[pba->index_bg_rho_cdm];
    p_tot += 0.;
    rho_m += pvecback[pba->index_bg_rho_cdm];
}

```